10/06/2026

Code-A Phase-1

Aakash

Medical | IIT-JEE | Foundations

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MM : 720

Fortnightly Test for NEET-2027 (XII-Passed)_FT01A

Time : 180 Min.

Topics Covered:
Physics: Units & Measurements, Motion in a Straight Line
Chemistry: Some Basic Concepts of Chemistry
Botany: Cell : The Unit of Life
Zoology: Structural Organisation in Animals– Animal Tissues Biomolecules-I: (Upto polysaccharides)

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

1. If $R = \frac{a^2 b^{\frac{1}{2}}}{c^3 d}$ and percentage error in measurement of a, b, c and d are 0.1%, 0.2%, 0.3% and 0.4% respectively, then the maximum percentage error in calculation of R is
(1) 1.9% (2) 3.1%
(3) 2.9% (4) 4.0%

2. If 86.45 is rounded off upto three significant figures then it should be written as
(1) 86.5 (2) 86.4
(3) 86.0 (4) 86.3

3. The resistance $R = \frac{V}{I}$ where $V = (80 \pm 4)$ V and $I = (10 \pm 0.1)$ A. The percentage error in R is
(1) 4% (2) 6%
(3) 8% (4) 9%

4. In a Vernier calliper 20 Vernier scale divisions are equal to 19 main scale divisions and each main scale division is of 1 mm. The least count of the Vernier calliper is
(1) 0.05 mm (2) 1 mm
(3) 0.5 mm (4) 2 mm

5. Which of the following is incorrect about significant figures?
(1) All non-zero digits are significant
(2) All zeroes appearing between two non-zero digits are significant
(3) The trailing zeroes in a number without decimal point are significant
(4) Both (1) & (2)

6. If the true value of time is 12:30:16 O'clock and two watches A and B show the time as written below. Select the single option in which watch A is most accurate and watch B is most precise
(1) A(12:30:10), B(12:30:16) (2) B(12:30:17), A(12:30:16)
(3) A(12:30:16), B(12:30) (4) B(12:30:16), A(12:30:12)

7. Time intervals measured by a clock gives the following readings:
1.25 s, 1.24 s, 1.27 s, 1.21 s and 1.28 s. What is the absolute mean error in the observations?
(1) 0.02 (2) 0.05
(3) 0.06 (4) 0.03

8. Time period of simple pendulum is given by $T = 2\pi\sqrt{\frac{l}{g}}$. if % error in time period is 2% and % error in length is 3%, find % error in acceleration due to gravity.
(1) 4% (2) 5%
(3) 6% (4) 7%

9. If the absolute errors in two physical quantities x and y are Δx and Δy respectively, then the maximum absolute error in the value of $x - y$ is
(1) $\Delta x - \Delta y$ (2) $\Delta x + \Delta y$
(3) $2\Delta x - \Delta y$ (4) $\Delta x - 2\Delta y$

10. Zero error in an instrument is a
(1) Random error (2) Systematic error
(3) Least count error (4) Personal error

11. If random error in arithmetic mean of 200 observations is e, then random error in arithmetic mean of 600 observations would be
(1) $4e$ (2) e
(3) $\frac{e}{3}$ (4) $\frac{e}{4}$

12. Consider the given statements
Statement A: Measured value by any measuring instrument has some error.
Statement B: A measurement can have more accuracy and less precision and vice-versa.
Statement C: The magnitude of the difference between the true value of the quantity and the individual measured value is called relative error of measurement.
Choose the correct option.
(1) Only statement A is true
(2) Only statement B is true
(3) Both statements A and C are true
(4) Both statements A and B are true

13. Consider the diameter of a spherical object being measured with the help of a Vernier callipers. Suppose its 10 Vernier Scale Divisions (V.S.D.) are equal to its 9 Main Scale Divisions (M.S.D.). The least division in the M.S. is 0.1 cm and the zero of V.S. is at $x = 0.1$ cm when the jaws of Vernier callipers are closed.
If the main scale reading for the diameter is $M = 5$ cm and the number of coinciding vernier division is 8, the measured diameter after zero error correction, is
(1) 5.08 cm (2) 4.98 cm
(3) 5.00 cm (4) 5.18 cm

14. Given below are two statements:
Statement-I: A choice of change in different units does not change the number of significant figures in a measurement.
Statement-II: when two physical quantities are multiplied, there should be as many decimal places retained in the final result as are there in the number with the least decimal places.
In light of above statements, choose the most appropriate answer from the options given below
(1) Both statements I and II are correct
(2) Both statements I and II are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

15. An expression of a dimensionless quantity $P = \frac{a}{b}\sin\left(\omega t + bx\right)$; where a and b are constants, x is distance and t is time. Then dimensions of a will be
(1) $[M^0L^{-1}T^{-1}]$ (2) $[M^0L^{-1}T]$
(3) $[M^0L^{-1}T^0]$ (4) $[M^0L^0T^{-1}]$

16. The potential energy of a particle varies with distance x from a fixed origin as $U = \left(\frac{A\sqrt{x}}{x+B}\right)$, where A and B are constants. The dimension of AB are
(1) $[ML^2T^{-2}]$ (2) $[ML^{7/2}T^{-2}]$
(3) $[ML^{5/2}T^{-2}]$ (4) $[ML^{7/2}T^{-3/2}]$

17. Which of the following pairs of physical quantities have same dimensional formula?
(1) Work and torque
(2) Angular momentum and Planck's constant
(3) Stress and Strain
(4) Both (1) and (2)

18. If y and x are pressure gradient and force respectively then dimension of $\frac{d^3y}{dx^3}$ equals.
(1) $[ML^{-5}T^{-4}]$ (2) $[M^{-2}L^{-5}T^4]$
(3) $[M^{-2}LT^{-4}]$ (4) $[M^{-2}L^2T^2]$

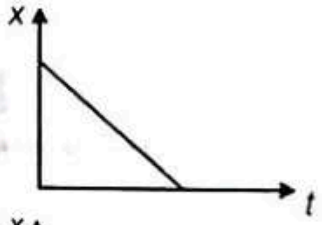
19. The numbers 2.735 and 2.785 on rounding off to three significant figures will be respectively
(1) 2.73 and 2.78 (2) 2.74 and 2.79
(3) 2.74 and 2.78 (4) 2.73 and 2.79

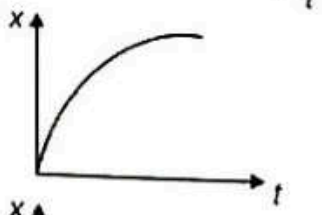
20. If mass (M), acceleration (A) and velocity (V) are taken as base quantities, then dimensional formula of energy is
(1) $[MV^{-2}A^4]$ (2) $[MV^2A^0]$
(3) $[M^{-1}V^3A^{-1}]$ (4) [MVA]

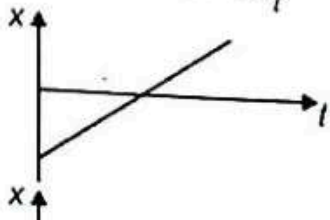
21. Match list-I with list-II

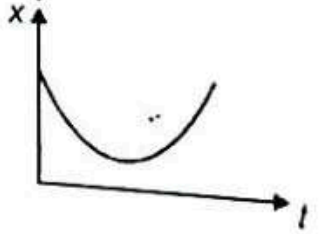
List-I	List-II
a. Velocity	(i) $[M^1L^1T^{-2}]$
b. Acceleration	(ii) $[M^0L^1T^{-2}]$
c. Force	(iii) $[M^1L^{-1}T^{-2}]$
d. Pressure	(iv) $[M^0L^1T^{-1}]$

Choose the correct answer from the options given below
(1) a(iv), b(ii), c(i), d(iii) (2) a(iv), b(ii), c(iii), d(i)
(3) a(iii), b(ii), c(i), d(iv) (4) a(ii), b(i), c(iii), d(iv)

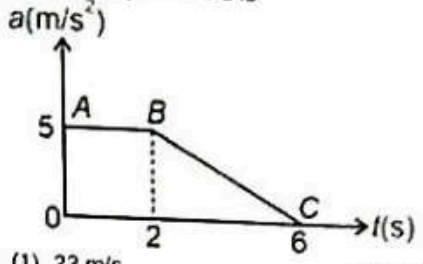
22. The position-time (x-t) graph showing negative acceleration is
(1)


(2)


(3)


(4)


23. A particle is moving on a straight line according to the given acceleration-time graph. If particle starts with velocity 3 m/s, then velocity at $t = 6$ s is



(1) 23 m/s
(2) 13 m/s
(3) 20 m/s
(4) 10 m/s

24. This section contains 1 Assertion-Reason type question, which has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.
A : A body can have velocity varying in both magnitude and direction but still have constant acceleration.
R : Acceleration is defined as rate of change of speed with respect to time.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

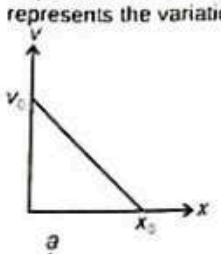
25. If velocity of a particle moving in straight line is given by $v = (5 + 2t)$ m/s, then average acceleration of particle between $t = 2$ s and $t = 5$ s is
(1) 2 m/s^2
(2) 3 m/s^2
(3) 0.5 m/s^2
(4) 4 m/s^2

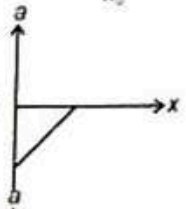
26. A man walks on a straight road from his home to a market 3 km away with a speed 6 km h^{-1}. Finding the market closed, he instantly turns and walks back with a speed 9 km h^{-1}. The magnitude of average velocity and average speed are respectively
(1) 2.25 km h^{-1}, 7.2 km h^{-1}
(2) 7.2 km h^{-1}, 2.25 km h^{-1}
(3) Zero, 2.25 km h^{-1}
(4) Zero, 7.2 km h^{-1}

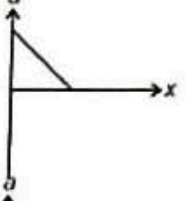
27. A train moving with a constant speed 72 km/h crosses a platform of length 400 m in 30 s. The length of train is
(1) 100 m
(2) 150 m
(3) 200 m
(4) 400 m

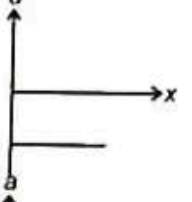
28. The value of $\int_0^{\pi} \sin x dx$ is
(1) 2
(2) –2
(3) 0
(4) 1

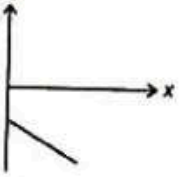
29. The given graph shows the variation of velocity with displacement. Which one of the graph given below correctly represents the variation of acceleration with displacement?



(1)


(2)


(3)


(4)


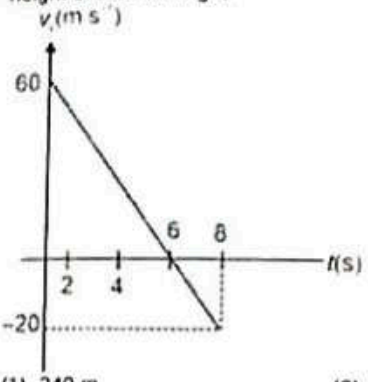
30. Position of a particle varies with time as $x = t^3 - 3t^2 + 4$. What will be its speed at the moment when its acceleration is zero? (x is in metre and t is in seconds)
(1) Zero
(2) 7 m/s
(3) 3 m/s
(4) 6 m/s

31. A train of length 200 m is going towards North at a speed of 15 m/s. A bird is flying at 5 m/s parallel to the track towards South. The time taken by the bird to cross the train is.
(1) 40 s
(2) 13.33 s
(3) 10 s
(4) 20 s

32. Two stones are thrown from top of a tower, one vertically upward and other downward with same speed. Ratio of their speeds when they hit the ground will be
(1) 1 : 4
(2) 1 : 1
(3) 1 : $\sqrt{2}$
(4) $\sqrt{2}$: 1

CC-533

33. The velocity-time graph for the vertical component of the velocity of a body thrown upwards from the ground and landing on the roof of a building is given in the figure. The height of the building is



(1) 240 m (2) 80 m
(3) 320 m (4) 160 m

34. Two trains having lengths 80 m and 120 m are running in the opposite directions with velocities 36 km/h and 54 km/h respectively. Time in which they will completely cross each other is

(1) 40 s (2) 12 s
(3) 15 s (4) 8 s

35. A person walked up a stalled escalator in 60 s. When standing on the same escalator, now moving, he is carried up in 30 s. How much time would it take him to walk up the moving escalator?

(1) 10 s (2) 90 s
(3) 30 s (4) 20 s

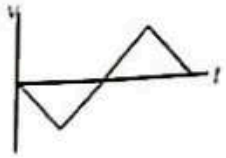
36. A body is dropped from the top of a tower. Distance covered by it in the last two seconds of its flight is equal to distance covered in first four seconds. Height of the tower is (g = 10 ms^{-2})

(1) 180 m (2) 160 m
(3) 105 m (4) 125 m

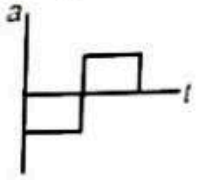
37. A ball is allowed to fall from top of a tower takes time t_1 to fall first half and t_2 in second half then $\left(\frac{t_2+t_1}{t_1}\right)$ is

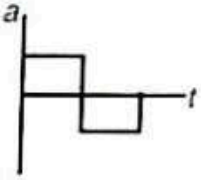
(1) $(\sqrt{2}-1)$ (2) $\frac{1}{(\sqrt{2}-1)}$
(3) $\frac{1}{\sqrt{2}}$ (4) $\sqrt{2}$

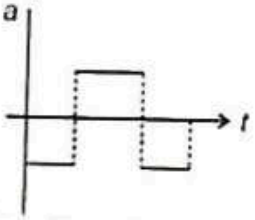
38. The velocity–time(v–t) graph for a particle in straight line motion is given below:

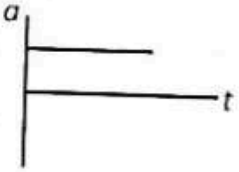


The corresponding acceleration-time graph is

(1)


(2)


(3)


(4)


39. A juggler throws ball into air. He throws one whenever the previous one is at its highest point. How high do the balls rise if he throws n balls each second?

(1) $\frac{g}{n^2}$ (2) $\frac{g}{2n^2}$
(3) $\frac{2g}{n^2}$ (4) $\frac{g}{n}$

40. Given below are two statements. One is labelled as Assertion (A) and other is Reason (R).

Assertion (A): If the velocity of a particle is zero at an instant its acceleration must be zero at that instant.

Reason(R): $a = \frac{dv}{dt}$ implies if $a = 0$, then $v = 0$.

In the light of the above statements, choose the most appropriate answer from the options given below:

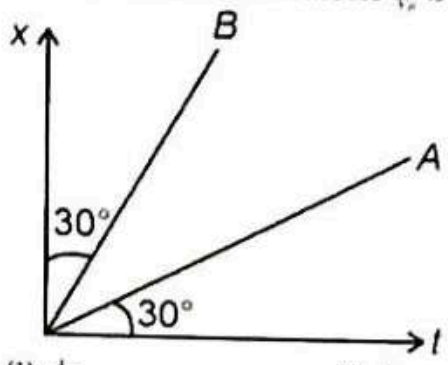
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

41. A particle moving with constant acceleration from A to B in the straight line AB has velocities u and v at A and B respectively. Find the velocity at C, the midpoint of AB if v = $7u$

(1) $3u$
(3) $2u$ (2) $5u$
(4) $4u$

42. A car starts from rest and accelerates at 1 ms^{-2} for 4 s. It then moves at a constant velocity for next 10 s, then decelerates at 2.0 ms^{-2} and finally comes to rest. The total distance travelled by the car is
(1) 52 m
(2) 40 m
(3) 48 m
(4) 46 m

43. The position – time graph for two cars *A* and *B* is given below, then the ratio of their velocities $\frac{v_A}{v_B}$ is



(1) $\frac{1}{\sqrt{3}}$
(2) $\frac{1}{1}$
(3) $\frac{\sqrt{3}}{1}$
(4) $\frac{2}{3}$

44. A police jeep is moving with 80 m/s, chasing a thief's car which is moving with 50 m/s. Police fires a bullet with muzzle velocity of 200 m s^{-1}, then the speed of the bullet as seen by the thief will be
(1) 330 m/s
(2) 230 m/s
(3) 150 m/s
(4) 200 m/s

45. Given below are two statements:
Statement-I : The velocity of an object can reverse its direction even when the acceleration of the object is constant.
Statement-II : It is possible that the acceleration of a body changes its direction without changing its direction of velocity.
In the light of the above statements, choose the most appropriate answer from the options given below.
(1) Both Statement-I and Statement-II are correct
(2) Both Statement-I and Statement-II are incorrect
(3) Statement-I is correct but statement-II is incorrect
(4) Statement-I is incorrect but statement-II is correct

CHEMISTRY

46. Consider the given reactions
(i) $N_2(g) + O_2(g) \rightarrow 2NO(g)$
(ii) $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$
Mass of NO_2 produced from 0.5 mol of N_2 is
(1) 92 g
(2) 72 g
(3) 69 g
(4) 46 g

47. Number of Fe atoms in 100 g Haemoglobin if it contains 0.33% Fe. (Atomic mass of Fe = 56 u)
(1) 0.035×10^{23}
(2) 35
(3) 3.5×10^{23}
(4) 7×10^{8}

48. The mass of 1.5055×10^{23} CO molecules is
(1) 14 g
(2) 7 g
(3) 3.5 g
(4) 28 g

49. Consider the following statements
Statement I: Equal masses of S_2 and S_8 contain same number of molecules.
Statement II: Equal masses of O_2 and O_3 contain equal number of atoms.
In the light of above statements, choose the correct option.
(1) Both the statements I and II are correct
(2) Both the statements I and II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

50. Percentage purity of 150 g of $CaCO_3$ sample which gives 5.6 litre CO_2 at STP on complete decomposition is
(1) 11.33%
(2) 16.67%
(3) 23.77%
(4) 33.33%

51. The number of electrons in 1.6 g of CH_4 is approximately
(1) 25×10^{24}
(2) 1.5×10^{24}
(3) 6×10^{23}
(4) 3.0×10^{24}

52. Read the following statements about $Ba_3(PO_4)_2$
a. Ratio of number of phosphorous to oxygen atoms is 3 : 4.
b. Ratio of number of cations to number of anions is 3 : 2.
c. Mass of oxygen required to produce 0.1 mole of $Ba_3(PO_4)_2$ is 12.8 g.
Choose the correct statement(s).
(1) a and b only
(2) a and c only
(3) b and c only
(4) c only

53. How many litre of oxygen at STP is required to burn 60 g C_2H_6?
(1) 22.4 L
(2) 11.2 L
(3) 22.4 x 7 L
(4) 8.5 L

54. In a one molal aqueous solution of a solute, the mole fraction of solvent is approximately
(1) 0.80
(2) 0.98
(3) 0.02
(4) 0.20

55. 34.2 g of aluminium sulphate is dissolved in enough water to make 500 mL of solution. If aluminium sulphate dissociates completely, then the molar concentration of Al^{3+} ions and SO_4^{2-} ions respectively are
(1) 0.4 M & 0.4 M
(2) 0.2 M & 0.4 M
(3) 0.4 M & 0.6 M
(4) 0.3 M & 0.6 M

56. If 14 g of nitrogen gas is mixed with 6 g of hydrogen gas, then the maximum mass of ammonia produced in the reaction is
(1) 17 g
(2) 15 g
(3) 20 g
(4) 12 g

57. 3.011×10^{21} molecules of glucose is present in 200 mL of its solution. The molarity of the solution is
(1) 2.5×10^{-2} M
(2) 2.5×10^{-3} M
(3) 4.5×10^{-1} M
(4) 4.5×10^{-2} M

58. The number of significant figures in 2.005 is
(1) 1 (2) 2
(3) 3 (4) 4

59. Consider the following statements
(a) 1 amu = 1.66056×10^{-27} kg
(b) 1 mole of atomic H contains 6.022×10^{-23} hydrogen atoms
(c) SI unit of luminous intensity is candela
The correct statements are
(1) (a) and (b) only (2) (b) and (c) only
(3) (a) and (c) only (4) (a), (b) and (c)

60. An element X has the following isotopic composition ^{60}X : 60% and ^{62}X 40%. The weighted average atomic mass of the naturally occurring element X is
(1) 60.4 amu (2) 60.5 amu
(3) 60.2 amu (4) 60.8 amu

61. Consider the given reaction
A + 2B → 3C
2 moles of each A and B have been taken in a closed container to form C.
Consider the following observations.
(a) A is limiting reagent
(b) 3 moles of C is formed
(c) 1 mole of B is left after the reaction.
The incorrect statement(s) is/are
(1) (b) only (2) (a) and (c) only
(3) (a) and (b) only (4) (b) and (c) only

62. Which of the following is the incorrect match?

	Prefix	Multiple
(1)	Milli	10^{-3}
(2)	Pico	10^{-12}
(3)	Nano	10^{-9}
(4)	Micro	10^{-4}

(1) (1) (2) (2)
(3) (3) (4) (4)

63. What is the molarity of H_2SO_4 solution containing 49% H_2SO_4 by mass? (density of solutions = 1.09 g/mL)
(1) 20 M (2) 5.45 M
(3) 15 M (4) 13.80 M

64. An organic compound contains carbon, hydrogen and oxygen. Its elemental analysis gave C = 40% and H = 6.67%. The empirical formula of the compound would be
(1) HCO (2) H_2CO
(3) HC_2O (4) H_2C_2O

65. 100 mL of 5 M NaOH solution was diluted by adding water to make the final solution having molarity 0.25 M. The volume of water added to the solution was
(1) 2.0 L (2) 1.6 L
(3) 1.9 L (4) 1.8 L

66. The number of hydrogen atoms present in 5.4 g of urea is:
(Given: Molar mass of urea : 60 g mol^{-1}, N_A : 6.022×10^{23} particles mol^{-1})
(1) 2.168×10^{23} (2) 2.168×10^{22}
(3) 1.084×10^{22} (4) 1.084×10^{23}

67. If 3.01×10^{23} molecules of sugar are present in 24.5 mol of water then mole fraction of sugar in solution is
(1) 0.08 (2) 0.04
(3) 0.2 (4) 0.02

68. Which of the following are correct statements?
(a) Precision refers to closeness of various measurements for the same quantity.
(b) Accuracy is the agreement of particular value to the true value of the result.
(c) Accuracy is closeness of various measurements for the same quantity.
(1) (a) & (b) only (2) (b) & (c) only
(3) (a) & (c) only (4) (a), (b) & (c)

69. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R :
Assertion A : CuO and Cu_2O illustrate the law of multiple proportion.
Reasons R : Percentage of copper and oxygen in samples of CuO obtained by different methods were found to be same.
In the light of the above statements, choose the correct answer from the options given below :
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

70. Consider the following statements
Statement-I: Gay Lussac's Law of Gaseous Volumes : When gases combine or are produced in a chemical reaction they do so in a simple ratio by volume provided that all gases are at same temperature and pressure.
Statement-II: The volumes of $H_2(g)$ and $O_2(g)$ which combine together to form water bear a simple ratio of 2 : 1
In the light of above statements, choose the correct answer from the options given below.
(1) Statement I is false but statement II is true
(2) Both statements I and II are true
(3) Both statements I and II are false
(4) Statement I is true but statement II is false

71. Maximum number of molecules are present in
(1) 15 L of H_2 gas at STP (2) 5 L of N_2 gas at STP
(3) 0.5 g of H_2 gas at STP (4) 10 g of O_2 gas at STP

72. If the mass ratio of He to N_2 in a container is 2 : 1 then the ratio of their moles will be
(1) 2 : 1 (2) 4 : 1
(3) 7 : 1 (4) 14 : 1

73. From 220 mg of CO_2, 10^{21} molecules of CO_2 are removed. Number of remaining molecules of CO_2 will be
(1) 3.011×10^{21} (2) 2.011×10^{21}
(3) 1.011×10^{21} (4) 6.022×10^{21}

74. A solution is prepared by adding 5 g of a solute in 15 g of water. The mass percent of the solute in the solution is
(1) 25% (2) 33.3%
(3) 50% (4) 10%

75. Consider the following reaction
$4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(l)$
If 2 mole of ammonia and 2 mole of oxygen are made to react to completion then choose the correct statement.
(1) Volume of excess reagent left is equal to 5.4 L at STP
(2) Moles of NO produced will be 2
(3) Molecules of H_2O produced is equal to 2.4 N_A
(4) Ammonia is the limiting reagent.

76. For the complete combustion of 44.8 L of propane (C_3H_8) at STP, the amount of oxygen gas (O_2) needed is
(1) 160 g (2) 320 g
(3) 64 g (4) 80 g

77. The mass of 3.01×10^{24} atoms of triatomic gas is 80 g. The molecular mass of gas is
(1) 240 (2) 40
(3) 26.67 (4) 48

78. A water bath has temperature 65°C. Its value in °F scale will be
(1) 142°F (2) 165°F
(3) 149°F (4) 155°F

79. The mass percentage of oxygen in H_2SO_4 is
(1) 15.23 (2) 65.31
(3) 17.15 (4) 23.24

80. Total number of electrons in 1 mol P^{3-} ions is
(1) 18 (2) 18 N_A
(3) 15 N_A (4) 10 N_A

81. The concentration of an aqueous solution of H_2SO_4 is 98% (w/w) and its density 1.8 g/ml. The volume of this acid required to make 500 ml of 0.2 M H_2SO_4 solution is
(1) 11.55 ml (2) 55.5 ml
(3) 25.5 ml (4) 5.55 ml

82. The formula of a compound is AB_3. If the mass of 0.25 moles of this compound is 66.75 g, the molecular mass of the compound is
(1) 137.5 u (2) 110 u
(3) 220 u (4) 267 u

83. The mole fraction of solute in aqueous solution containing 42.5% (w/w) $NaNO_3$ is
(1) 0.251 (2) 0.053
(3) 0.215 (4) 0.135

84. The concentration of aq. 40% (w/v) glucose solution in ppm is (density of solution is 1.2 g/ml)
(1) 3.3×10^5 ppm (2) 4.5×10^4 ppm
(3) 5.5×10^6 ppm (4) 1.5×10^8 ppm

85. A gaseous mixture contains H_2 and CO_2 gases in the ratio of 1 : 11 (w/w). The ratio of number of atoms of H_2 to CO_2 in the mixture is
(1) $\frac{4}{3}$ (2) $\frac{2}{5}$
(3) $\frac{1}{3}$ (4) $\frac{1}{7}$

86. Match the given columns

Column I (Composition)	Column II (Molality)
a. 18 g glucose ($C_6H_{12}O_6$) in 180 g water	(i) $\frac{25}{9}\ m$
b. 34.2 g sucrose ($C_{12}H_{22}O_{11}$) in 360 g water	(ii) $\frac{50}{9}\ m$
c. 5.85 g of sodium chloride (NaCl) in 18 g of water	(iii) $\frac{5}{18}\ m$
d. 4.6 g ethyl alcohol (C_2H_5OH) in 36 g water	(iv) $\frac{5}{9}\ m$

Choose the correct option.
(1) a(ii), b(iv), c(i), d(iii) (2) a(ii), b(i), c(iv), d(iii)
(3) a(iv), b(iii), c(ii), d(i) (4) a(iv), b(iii), c(i), d(ii)

87. Given below are two statements:
Statement I : Molecular formula mass of benzene is six times of its empirical formula mass.
Statement II : An empirical formula represents the simplest whole number ratio of various atoms present in a compound.
In the light of above statements, choose the most appropriate answer from the options given below.
(1) Statement I is incorrect but statement II is correct.
(2) Statement I is correct but statement II is incorrect.
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect.

88. Suppose the elements X and Y combine to form two compounds X_2Y and X_3Y_4. 0.1 mole of X_2Y weighs 10 g and 0.05 mole of X_3Y_4 weighs 15 g, the atomic masses of X and Y respectively are
(1) 20 u and 60 u (2) 50 u and 30 u
(3) 60 u and 20 u (4) 30 u and 50 u

89. Incorrect statement regarding Dalton's Atomic theory is
(1) Matter consists of indivisible atoms
(2) All the atoms of a given element have identical properties
(3) Compounds are formed when atoms of same elements combined in a fixed ratio
(4) Chemical reactions involve reorganisation of atoms

90. If percentage of Zn (atomic mass = 65.38 u) in a biomolecule is 0.75% by weight, then the minimum molar mass (in g mol^{-1}) of biomolecule will be
(1) 10243.45 (2) 9483.29
(3) 8012.12 (4) 8717.33

BOTANY

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91. Match the following columns and select the **correct** option.

	Column I		Column II
(a)	Polysome	(i)	Single membrane bound minute vesicles containing enzymes
(b)	Lysosome	(ii)	Several granular structures that are composed of rRNA and proteins attached to a single mRNA forming a chain and can be observed in prokaryotes
(c)	Mesosome	(iii)	Membrane bound vesicular structure considered as a part of endomembrane system
(d)	Peroxisome	(iv)	A special membranous structure which is formed by the extensions of plasma membrane.

(1) a(ii), b(iii), c(i), d(iv)
(2) a(ii), b(iii), c(iv), d(i)
(3) a(iii), b(ii), c(iv), d(i)
(4) a(ii), b(iv), c(i), d(iii)

92. Read the following statements and choose the option for **correct** one(s).
A. Each layer of the cell envelope in bacteria performs distinct function.
B. Bacteria are classified into Gram positive and Gram negative on the basis of their response to a particular stain.
C. The composition of glycocalyx in all types of bacteria is same.
D. In some bacteria, cell wall forms thick and tough structure called capsule.
E. The plasma membrane in bacteria is selectively permeable in nature and interacts with the outside world.
(1) A and E only
(2) B, C and D
(3) A, B and E
(4) B only

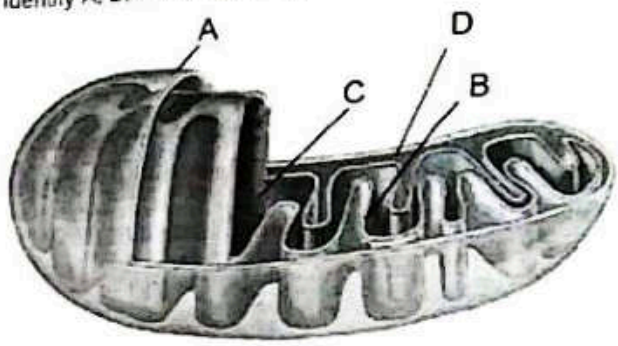
93. In SAT chromosome, a small fragment of chromosome separated by a non-stained secondary constriction, is called
(1) Centromere
(2) Chromatid
(3) Kinetochore
(4) Satellite

94. 80S ribosome has the subunits
(1) 60S and 40S
(2) 50S and 30S
(3) 16S and 23S
(4) 60S and 30S

95. Sarcoplasmic reticulum
(1) Is site of carbohydrate synthesis
(2) Has ribophorin I and II
(3) Is SER of muscle cells
(4) Is involved in the formation of enzymes of lysosomes

96. A polymorphic cell organelle, which is bound by a single membrane, is **not** associated with
(1) Digestion of food materials
(2) Conversion of fats into carbohydrates
(3) Lysis of old cell organelles
(4) Ephagy of undigested substances

97. Identify A, B, C and D in the given diagram.



(1) A – Inner membrane; B – Thylakoids
(2) B – Crista; C – Matrix
(3) C – Inter-membrane space; D – Cisternae
(4) D – Matrix; A – Tonoplast

98. Read the following statements and mark them as true (T) or false (F) and choose the correct option
A. In a eukaryotic cell, there is no organelle which lacks DNA but is capable of duplication.
B. 9 + 2 organisation of microtubules is found in the axoneme of basal bodies and centrioles.
C. Cell is the fundamental, structural and functional unit of all living organisms because anything less than a complete structure of a cell can ensure independent living.
D. The function of polysome in bacterial cell is to translate mRNA into proteins.
E. Eukaryotic cells have a well organized nucleus and their DNA is complexed with histones to constitute the chromatin.

	A	B	C	D	E
(1)	F	T	F	T	T
(2)	F	F	F	T	T
(3)	T	F	F	T	F
(4)	F	T	T	F	T

(1) (1)
(2) (2)
(3) (3)
(4) (4)

99. Small bristle like fibres sprouting out of the bacterial cells are helpful in
(1) Cell wall formation
(2) Attachment to the host tissue
(3) Secretion process
(4) Cell motility

100. The chromosomes that have satellite
(1) Lack primary constriction
(2) Have secondary constriction
(3) Do not undergo replication
(4) Are always telocentric

101. Organelle in eukaryotic cell that has an organisation like a cartwheel
(1) Has peripheral doublets
(2) Is surrounded by membrane
(3) Is surrounded by amorphous pericentriolar materials
(4) Occurs in pair lying parallel to each other

102. Which of the following statements regarding chromatophores is **incorrect**?
(1) These are the membranous extensions into the cell
(2) These are found in all photosynthetic organisms
(3) They have photosynthetic pigments
(4) These structures have lipids and proteins

103. Ribosomal subunits remain united with each other due to a specific concentration of
(1) Ca^{2+}
(2) Mg^{2+}
(3) Mn^{2+}
(4) Fe^{3+}

104. Which of the following is **false** about lysosomes?
(1) Single membrane bound organelle
(2) Very rich in hydrolytic enzymes
(3) Enzymes optimally active at basic pH
(4) Formed by Golgi apparatus

105. Match the column I with column II and select the **correct** option.

	Column I		Column II
a.	Elaioplast	(i)	Stores starch
b.	Aleuroplast	(ii)	Contains carotenoid
c.	Amyloplast	(iii)	Stores oils and fats
d.	Chromoplast	(iv)	Stores proteins

(1) a(ii), b(i), c(iv), d(iii)
(2) a(i), b(iv), c(iii), d(ii)
(3) a(iii), b(i), c(iv), d(ii)
(4) a(iii), b(iv), c(i), d(ii)

106. The site of oxidative phosphorylation is
(1) Chloroplast
(2) Mitochondria
(3) Ribosomes
(4) Leucoplast

107. Theodore Schwann first reported thin outer layer, *i.e.*, plasma membrane in
(1) Plant cells
(2) Animal cells
(3) Bacterial cells
(4) Algal cells

108. Given below are two statements:
Statement I: Primary wall of the plant cell shows growth even after the cell becomes mature.
Statement II: Cell wall protects the cell from infection and it also helps in cell-to-cell interaction.
In the light of above statements, choose the **correct** answer from the option given below:
(1) Both statement I and statement II are true
(2) Both statement I and statement II are false
(3) Statement I is true and statement II is false
(4) Statement I is false and statement II is true

109. Select the **Incorrect** match from the following.
(1) Anthocyanin – Water soluble pigment
(2) Cytoskeleton – Proteinaceous structures
(3) Chloroplast – Autonomous organelle
(4) Protoplast – Cell without cell wall

110. Reserve material in prokaryotic cells
(1) Are stored in the nucleus of the cell
(2) Are bound by double membrane
(3) Are usually found in the form of starch granules inside the vacuoles
(4) Lie free in the cytoplasm

111. For the first time, Rudolf Virchow told that *Omnis cellula-e cellula*. It means
(1) All living organisms are composed of cells
(2) Each cell has organelles that function in coordinated manner
(3) All cells arise from pre-existing cells
(4) Activities of an organism are the result of activities of its constituent cells

112. Read the following **Assertion (A)** and **Reason (R)** and select the **correct** option.
Assertion (A) : In prokaryotes, ribosomes are associated with plasma membrane of the cell.
Reason (R) : Only one type of ribosomes having 40S and 30S subunits are found in prokaryotes.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

113. The function which is **not** associated with microtubules is
(1) Formation of spindles and astral rays
(2) Formation of basal body of flagella in eukaryotes
(3) Help in formation of cleavage furrow during cell division
(4) Help in formation of cell plate during plant cell cytokinesis

114. Select the odd **one** w.r.t prokaryotic cells
(1) PPLO
(2) Yeast
(3) Blue green algae
(4) Gram negative bacteria

115. Endomembrane system includes all, **except**
(1) ER
(2) Golgi complex
(3) Lysosome
(4) Gas vacuole

116. A disc shaped structure on sides of centromere is called
(1) Centrosome
(2) Chromatid
(3) Kinetochore
(4) Telomere

117. Spindle fibres are made up of protein
(1) Actin
(2) Flagellin
(3) Myosin
(4) Tubulin

118. Camillo Golgi first observed densely stained reticular structures near the nucleus. These structures have
a. Many flat, disc-shaped sacs of 0.5 µm to 1.0 µm in diameter.
b. *Cis* and *trans* faces that are called maturing face and forming face, respectively.
c. Certain functions like packaging materials and transport of the packed material to either intra-cellular targets or outside of the cell.
d. Close association with the organelles of endomembrane system, such as peroxisomes.
The **correct** ones are
(1) Only a, b and c
(2) Only a, b and d
(3) Only b, c and d
(4) Only a and c

119. Spherical structures present in nuclear matrix and are site of active r-RNA synthesis are **absent** in
(1) *Chlamydomonas*
(2) *Amoeba*
(3) Purple and green photosynthetic bacteria
(4) A plant cell

120. Match the following **List I** with **List II**

	List-I		List-II
A.	Sub-metacentric chromosome	(i)	Has one extremely short and one very long arm
B.	Acrocentric chromosome	(ii)	p and q arms are indistinguishable
C.	Telocentric chromosome	(iii)	Centromere is slightly away from the middle of chromosome
D.	Metacentric chromosome	(iv)	It has terminal centromere

Choose the **correct** answer from the options given below

(1) A-(iii), B-(iv), C-(i), D-(ii) (2) A-(ii), B-(iii), C-(iv), D-(i)
(3) A-(iv), B-(iii), C-(i), D-(ii) (4) A-(iii), B-(i), C-(iv), D-(ii)

121. State true **(T)** or false **(F)** for the given statements and choose the **correct** option

a. Na^+/K^+ pump in animal cells is a membrane protein involved in uphill movement of ions.
b. Cell growth and formation of intercellular junctions are functions which are feasible due to non-fluid nature of plasma membrane.
c. During endocytosis, exudation of cellular materials in the form of carrier vesicle occurs through plasma membrane.
d. Water may also move across the plasma membrane from its higher to lower concentration.

	a	b	c	d
(1)	F	T	T	F
(2)	T	F	F	T
(3)	T	T	F	T
(4)	F	F	T	F

(1) (1) (2) (2)
(3) (3) (4) (4)

122. Given below are two statements:
Statement I: Some bacteria are resistant to antibiotics due to presence of an additional small circular DNA.
Statement II: Single stranded linear DNA confers unique phenotypic characters to most of the bacteria.
In the light of above statements, choose the **correct** answer from the option given below:
(1) Both statement I and statement II are true
(2) Both statement I and statement II are false
(3) Statement I is true and statement II is false
(4) Statement I is false and statement II is true

123. Observe the following steps w.r.t. process, package and transport of materials by endomembrane system.
a. Movement of biomolecules from ER to Golgi apparatus in the form of transitional vesicles.
b. Release of vesicles from concave face of Golgi apparatus.
c. Synthesis of biomolecules in the network of tiny tubular structures scattered in the cytoplasm.
d. Movement of vesicle from forming to maturing face.
Select the option that depicts the correct chronological sequence of the given events.
(1) a → b → d → c (2) c → a → d → b
(3) b → c → a → d (4) c → b → d → a

124. The flat membranous tubule like structure which connects the thylakoids in chloroplast is called
(1) Grana (2) Stroma lamellae
(3) Lumen (4) Stroma

125. Why chloroplast is not considered as part of the endomembrane system?
(1) It is not involved in oxidation of fats
(2) It is double membrane bound structure
(3) Its functions are not co-ordinated with the functions of cell organelles of endomembrane system
(4) It stores starch

126. The inner membrane of mitochondria
(1) Is more permeable than the outer membrane
(2) Consists of 40% lipids and 60% proteins
(3) Is rich in cardiolipins
(4) Is smooth and forms the continuous limiting boundary

127. The main arena of cellular activities in plant and animal cell is
(1) Nucleoplasm (2) Cytoplasm
(3) Mitochondrial matrix (4) Perimitochondrial space

128. Chromatin contains
(1) DNA and RNA only
(2) DNA and some basic proteins only
(3) DNA, histones, non-histones proteins and RNA
(4) Only DNA

129. Read the following **Assertion (A)** and **Reason (R)** statements and select the correct option.
Assertion (A): A phospholipid bilayer can exist as a stable boundary between two aqueous compartments.
Reason (R): The hydrophilic head of phospholipid bilayer protects the hydrophobic tails from aqueous environment.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) Both (A) and (R) are false.

130. Centrioles are found in almost all eukaryotic cells, **except**
(1) Animal cells (2) Fungal cells
(3) Algal cells (4) Higher plants cells

131. Regarding microbodies of the cells identify the option that is NOT true.
(1) They contain various enzymes
(2) These are present in cells of both photosynthetic and non-photosynthetic organisms
(3) Their inner membrane forms infoldings for enzyme actions
(4) These are in the form of minute vesicles

132. Select the **incorrect** statement.
(1) Flagella are comparatively longer than cilia & responsible for cell movement
(2) The prokaryotic flagella are functionally different from eukaryotic flagella
(3) The axoneme of cilia usually has nine peripheral doublets of microtubules
(4) Centrioles form the basal body of eukaryotic cilia and flagella

133. Nucleus as a cell organelle was first described by
(1) Theodor Schwann (2) Robert Brown
(3) Robert Hooke (4) Matthias Schleiden

134. The shape of mesophyll cells is primarily
(1) Amoeboid (2) Elongated
(3) Branched and long (4) Round and oval

135. Middle lamella of plant cell
(1) Is composed of cellulose only
(2) Is composed of galactans and mannans
(3) Holds the two neighbouring cells together
(4) Is composed of hemicellulose mainly

136. Choose the **incorrect** statement
(1) Filtrate and retentate are two fractions obtained during chemical analysis of a living tissue
(2) Inorganic compounds like sulphates and phosphates are found in acid-insoluble fraction
(3) All the organic compounds that we get from living tissues can be called biomolecules
(4) Oxygen constitutes the maximum percentage weight of Earth's crust as well as the human body

137. The type of tissue which faces either the body fluid or the outside environment and thus provides lining or covering for the hollow visceral organs is
(1) Epithelial tissue
(2) Connective tissue
(3) Muscular tissue
(4) Nervous tissue

138. All of the following cells are present in areolar tissue, **except**
(1) Fibroblasts
(2) Macrophages
(3) Mast cells
(4) Goblet cells

139. 'X' and 'Y' are two different types of connective tissues. 'X' is responsible for attaching one bone to another and 'Y' is designed specifically to store fats. Select the option that **correctly** identifies 'X' and 'Y' respectively.
(1) Ligament, adipose tissue
(2) Areolar tissue, tendon
(3) Tendon, adipose tissue
(4) Cartilage, dense connective tissue

140. The type of epithelium lining the proximal convoluted tubule is characterised by all of the following, **except**
(1) Performs secretion and absorption
(2) Possesses tall cells with nuclei located at the base
(3) Has microvilli on the free surface
(4) Has a single layer of cube-like cells

141. All of the following are the characteristic features of epithelium forming lining of alveoli, **except**
(1) A single thin layer of flattened cells with irregular boundaries
(2) Cells are compactly packed with little intercellular matrix
(3) The cells are ciliated
(4) The nuclei of cells are flat and often lie at the centre of cells

142. Common feature of starch and glycogen is that both
(1) Are structural polysaccharides
(2) Are storage polysaccharides
(3) Are heteropolymers of glucose and fructose
(4) Give red colour with I_2 solution

143. Choose the correct option to fill in the blanks respectively.
The tissues are organised in _____ and _____ to form an organ.
(1) Variable proportion; function
(2) Specific proportion; pattern
(3) Specific proportion; variable function
(4) Variable proportion; origination

144. Select the **incorrect** statement w.r.t. epithelial cells.
(1) Cells in epithelium are held together with little intercellular material
(2) Cell junctions are found in the epithelial tissue
(3) Epithelium represents only one free surface facing body fluid or outside environment
(4) The epithelium covering skin has diffusion, absorption, excretion and filtration function

145. The element which forms 18.5% weight of human body is
(1) Hydrogen
(2) Carbon
(3) Oxygen
(4) Calcium

ZOOLOGY

146. The exoskeleton of an arthropod is made up of a biomacromolecule, which is a polymer of
(1) N-acetyl glucosamine and N-acetyl muramic acid
(2) N-acetyl glucosamine
(3) N-acetyl muramic acid
(4) N-acetyl galactosamine and N-acetyl muramic acid

147. Dense connective tissues consist of compactly packed fibres and fibroblasts and it can be further categorised into dense regular and dense irregular tissues on the basis of
(1) Orientation of fibres
(2) Origin of fibres
(3) Number of fibres
(4) Structural composition of fibres

148. Find the **correct** match.

Column - I (Group)		Column - II (Secondary metabolite examples)
a. Pigment	(i)	Morphine
b. Alkaloid	(ii)	Ricin
c. Toxin	(iii)	Rubber
d. Polymer	(iv)	Monoterpene
	(v)	Anthocyanin

(1) a(i), b(v), c(iv), d(iii)
(2) a(i), b(iii), c(ii), d(iv)
(3) a(i), b(iv), c(ii), d(iii)
(4) a(v), b(i), c(ii), d(iii)

149. A histologist observes an epithelium containing single layer of tall and slender cells, having basal nuclei with mucous secreting cell in between them, on a slide through microscope.
Epithelium being discussed in the above statement is
(1) Simple squamous epithelium
(2) Simple cuboidal epithelium
(3) Stratified squamous epithelium
(4) Columnar glandular epithelium

150. An organ such as heart comprises
(a) Muscular tissue
(b) Neural tissue
(c) Connective tissue
(d) Epithelial tissue
Select the correct option.
(1) (a) & (b) only
(2) (a) only
(3) (b), (c) and (d) only
(4) (a), (b), (c) & (d)

151. The **incorrect** statement w.r.t. primary metabolites is that they
(1) Are required for basic metabolic processes
(2) Have identifiable functions
(3) Do not take part in physiological processes
(4) Include nucleic acids

152. The biomolecule that **cannot** be hydrolysed further is
(1) Starch
(2) Glycogen
(3) Cellulose
(4) Glucose

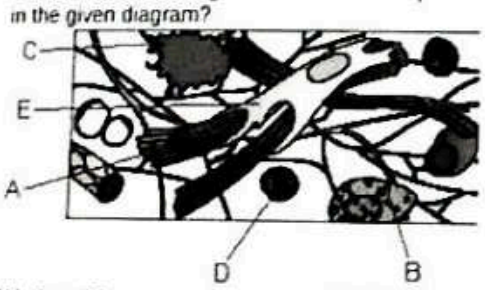
153. All of the following are true for muscle fibres present in the wall of stomach, **except**
(1) Fusiform in shape
(2) Branched muscle fibres
(3) Involuntary in nature
(4) Absence of striations

154. **Assertion (A)** : Out of all the types of connective tissues, blood is the main circulating fluid that helps in the transport of various substances across the body.
Reason (R) : Blood is a loose connective tissue in which fibroblasts are absent.
In the light of above statements, select the correct option.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) (A) is true but (R) is false
(3) Both (A) and (R) are false
(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

155. The relative abundance of _____ and _____ is higher in living organisms as compared to Earth's crust.
Select the correct option to fill in the blanks respectively.
(1) Carbon, hydrogen
(2) Carbon, calcium
(3) Oxygen, sodium
(4) Hydrogen, silicon

156. Choose the option which represents the correct decreasing order of cellular components according to their percentage of the total cellular mass.
(1) Water > Proteins > Nucleic acids > Carbohydrates > Lipids > Ions
(2) Nucleic acids > Proteins > Lipids > Carbohydrates > Ions > Water
(3) Proteins > Nucleic acids > Ions > Carbohydrates > Water > Lipids
(4) Ions > Lipids > Carbohydrates > Nucleic acids > Proteins > Water

157. Which of the following labelled structures produce histamine in the given diagram?



(1) A and B
(2) E only
(3) B only
(4) C and D

158. The retentate obtained after straining the slurry made by grinding a living tissue in trichloroacetic acid contains which set of compounds?
(1) Proteins, minerals, lipids
(2) Phosphate ions, sulphate ions, magnesium ions
(3) Phosphate ions, sulphate ions, proteins
(4) Proteins, nucleic acids, polysaccharides

159. How many of the secretions mentioned in the box given below are produced from exocrine glands?

Oil, Mucus, Wax, Saliva, Insulin, Thyroxine

Select the **correct** option.
(1) Four
(2) Three
(3) Five
(4) Two

160. Consider the following statements w.r.t. secondary metabolites.
a. Concanavalin A is a lectin.
b. Some secondary metabolites have ecological importance.
c. Cellulose is an alkaloid.
d. Codeine is a pigment.
Choose the option which includes all **correct** statements.
(1) Only a and b
(2) Only a, b and d
(3) Only b, c and d
(4) a, b, c and d

161. Complete the analogy.
Multicellular gland : Salivary gland :: Unicellular gland : _____
Choose the **correct** option.
(1) Goblet cell
(2) Sebaceous gland
(3) Sweat gland
(4) Gastric gland

162. The type of epithelium present in the inner lining of ducts of salivary gland, is also present in the
(1) Walls of blood vessels
(2) Lining of stomach and intestine
(3) Inner surface of hollow organs like bronchioles
(4) Moist surface of buccal cavity

163. Most abundant structural polysaccharide in nature is a secondary metabolite, named as
(1) Glycogen
(2) Starch
(3) Insulin
(4) Cellulose

164. Inulin is a polymer of
(1) Glucose
(2) Fructose
(3) Galactose
(4) Lactose

165. How many of the structures mentioned in the box below are included in the broad class of tissues which is most abundant and widely distributed in the body of complex animals?

Cartilage, Aerolar tissue, Blood, Adipose tissue, Bone

Choose the **correct** option.
(1) Two
(2) Three
(3) Four
(4) Five

166. 10–15% of total cellular mass is composed of
(1) Nucleic acids
(2) Carbohydrates
(3) Lipids
(4) Proteins

167. Skeletal muscle fibres are
(1) Branched, spindle shaped and uninucleated
(2) Striped, multinucleated and unbranched
(3) Unbranched, involuntary and cylindrical in shape
(4) Voluntary, unbranched and uninucleated

168. Which tissue exerts the greatest control over the body's responsiveness to changing conditions?
(1) Muscle tissue
(2) Neural tissue
(3) Connective tissue
(4) Epithelial tissue

169. ___A___ facilitates the cells to communicate with each other by connecting the cytoplasm of adjoining cells. Choose the option that **correctly** fills the blank.
(1) Tight junction
(2) Adhering junction
(3) Gap junction
(4) Desmosomes

170. Read the following given statements and choose the **correct** option.
Statement A : All the carbon compounds that we get from living tissues can be called biomolecules.
Statement B : Living organisms do not possess inorganic elements.
(1) Both statements are correct
(2) Both statements are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

171. What do fibroblast, chondroblast, osteoblast have in common?
(1) They are all types of white blood cells
(2) They all are mature cells
(3) They all are immature cells
(4) They all are types of epithelial cells

172. Read the following statements w.r.t. cartilage and choose the **correct** option.
Statement A : The matrix of cartilage is solid, pliable and resists compression.
Statement B : Osteocytes lie in lacunae in groups of two or four.
(1) Both statements are correct
(2) Both statements are incorrect
(3) Statement A is incorrect but B is correct
(4) Statement A is correct but B is incorrect

173. The innermost layer of walls of blood vessels is composed of
(1) Ciliated columnar epithelium
(2) Columnar epithelium
(3) Cuboidal epithelium
(4) Squamous epithelium

174. Select the **correct** statements w.r.t. bones.
(a) They support and protect softer tissues and organs.
(b) Limb bones serve weight bearing functions.
(c) Their interaction with muscles having fusiform shaped fibres bring about movements.
(d) All bones are site of production of blood cells in aged individuals.
Choose the most appropriate answer from the options given below.
(1) (a) and (b)
(2) (a) and (c)
(3) (c) and (d)
(4) (b) and (d)

175. Read the following statements and select the **correct** option.
Statement I: In the structure of a glycogen molecule, a free aldehyde group is present at left end of polysaccharide chain.
Statement II: Starch and cellulose can be distinguished on the basis of iodine test as cellulose gives blue colour and starch gives red colour with iodine.
(1) Both statements I and II are correct
(2) Only statement I is correct
(3) Only statement II is correct
(4) Both statements are I and II incorrect

176. Properties shared by both neural and muscular tissue usually include
(1) Elasticity
(2) Excitability
(3) Extensibility
(4) Germ layer of their origin

177. Category of storage polysaccharides does **not** include
(1) Inulin
(2) Chitin
(3) Glycogen
(4) Starch

178. Neuroglia
a. Protect neurons
b. Support neurons
c. Make up more than one-half the volume of neural tissue
Choose the option with the **correct** set.
(1) Only a and b
(2) Only a and c
(3) Only b and c
(4) a, b and c

179. The ash obtained from a tissue may contain how many of the following elements?

Sodium, Potassium, Calcium, Magnesium; Carbon, Oxygen, Hydrogen

(1) 7
(2) 6
(3) 3
(4) 4

180. Striated muscles fibres are responsible for muscle contraction in all of the following, **except**
(1) Running muscles in an athlete
(2) Digestive tract muscles in an obese man
(3) Fighting muscles of a boxer
(4) Cardiac muscles of a hunter